

The Sense of Agency Improves Visual Search and Holds Up Against External Salient Distractions in Virtual Reality

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ABSTRACT

Previous research has demonstrated that individuals are able to process information more quickly when they feel in control of the stimuli they are interacting with. However, it was unclear how this effect interacts with other visually salient information. To investigate this issue, we invited 39 college students to control hand-shaped object in a virtual environment while the remaining three objects displayed pre-recorded rotational movements. Following this task, we asked participants to identify a hand-shaped object with thumb pointing upward, which was the designated search target. Irrespective of the validity of the visually salient attentional cue, participants responded 11% faster when the search target corresponded with the previously controlled target. We concluded that the facilitation effect of a sense of agency in visual searching is robust to external distractors. Embracing the resilience of this facilitation effect arising from a sense of agency in visual search tasks can offer valuable insights for enhancing the design of diverse systems and environments to optimize human performance and experience.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Laboratory experiments.

KEYWORDS

Sense of Agency, Visual search, Salient attentional cue, Virtual Reality

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1 INTRODUCTION

Sense of agency refers to the perception of being in control of one's own actions and their effects on external world [1]. This perception arises from a variety of behaviors involving voluntary actions and their resulting effects [2]. For example, in our daily routines, we may experience a sense of agency when we press a button to turn on the television and feel a sense of accomplishment after successfully conducting this action. Moreover, in recent years, increasing number of studies have delved into the neural mechanisms underlying the sense of agency [3–6].

Researchers have explored “What is” and “Why there is” sense of agency from various perspectives. Health psychology, for instance, characterizes sense of agency as the capacity to persevere in pursuing one's professional and lifelong aspirations [7]. Yet, in the realm of engaging with media and technology, people are intrigued in understanding about “How” this cognitive factor might influence the perception of virtual environment, such as virtual reality, augmented reality, and video games.

In human perception, visual attention plays a crucial role in processing information from the external world, as individuals need to efficiently identify self-relevant information in a complex and rich environment. Previous studies have explored the relationship between agency and visual attention, with findings indicating that a sense of agency can guide the attention toward controlled stimulus [8], and that voluntary control, as opposed to involuntary control, enhances the recognition of the avatar representing oneself [9].

In addition to attention, noise serves as another significant variable that can influence individuals' processing of external stimuli. In certain circumstances, environmental noise may disrupt the effective perception of external information, thereby diminishing one's sense of agency in specific tasks. For instance, stormy weather interferes with the accurate perception and estimation of traveling speed of vehicles. An individual attempting to respond to messages on their mobile devices may become distracted by pop-up notifications from social media and system updates. Research has shown that the sense of agency is compromised in the presence of exogenous interference or visually salient distractions [10].

Expanding on the research conducted by Vilanova-Goldstein and his team [10], our study aims to investigate the robustness of sense of agency in the presence of visual distractions in a virtual environment. In comparison to their experiment, we have introduced a distinction between controlled and uncontrolled stimuli based on the source of control. The uncontrolled stimuli, which were prerecorded, do not share the same onset for the rotational movement with the stimulus controlled by participant. To create a more

immersive experimental environment for participants, we have implemented the study in a virtual reality set up. Participants interact with a virtual hand model instead of observing their own hands, facilitating a fully agency-driven experiment that minimizes the influence of visual inconsistency and other potential confounding variables.

In this study, we are particularly interested in exploring the interaction between sense of agency and visually striking external stimuli on visual search performance. The research questions guiding our study are as follows:

RQ1. Can the manipulation of prior control over a target result in expedited detection of said target during visual search in a virtual environment?

RQ2. Given that salient cues possess the ability to attract attention pre-attentively, to what extent and in what manner does the sense of agency function as a barrier against exogenous interference caused by salient attentional distractions?

2 RELATED WORK

In this section, we have provided a concise overview of the current theories on the sense of agency. Following that, we report the recent laboratory studies that examined how the sense of agency impacts the dynamics of visual search.

2.1 Sense of Agency in Cognition and Perception

When individuals engage in an activity and perceive that they have control over both their actions and the outcomes, this subjective experience is known as the sense of agency [11]. The sense of agency is intricately linked to actions and their effects, involving the motor system, perceptual system, and sensorimotor system in its formation [12]. Moreover, the sense of agency can be seen as an internal reward for voluntary motor control, rather than a passive byproduct [13, 14]. It could directly influence the motor control and enhance perception abilities in various ways [7, 15].

2.2 Sense of Agency in Visual Search

The study conducted by Wen and Haggard integrated the sense of agency induction with a visual selection task, where participants were required to choose one circle with a single gap from a set of circles with two gaps [7]. By manipulating the controllability of the visual search target, specifically the proportion of the actual trajectory controlled by the participants using a mouse, while keeping the controllability of distractors constant, they observed that the visual search target only captured more attention when its controllability surpassed that of the distractors.

Expanding on this research, Vilanova-Goldstein and his team made certain adaptations to the experimental design in examining the effects of sense of agency on visual search [10]. They introduced diverse types of attentional cues and shifted the control method from a cursor to key press. Their findings indicated that the sense of agency had a positive effect on visual attention, particularly in the presence of endogenous attentional cues.

2.3 Expectation and Statistical Learning in Visual Search

Regarding visual search, previous research has indicated that individuals tend to gravitate toward patterns and regularities [16, 17]. Across different experiments, participants consistently showed a preference for directing their focal attention towards areas of a display that had a greater likelihood of containing the visual search target. Scholars have recognized and highlighted the probabilistic learning as a key influence in target selection, alongside physical attributes and goal-relevance [18].

2.4 Exogenous Attentional Cue in Visual Search

In addition to expectation (i.e., top-down attention), attention can also be drawn to salient external information during the visual search, such as the stimuli with distinct physical properties such as a bright color and a sharp contrast. In a classic psychophysics experiment, participants were instructed to search for green circles among red squares. Although the distracting red squares shared no visual similarities with the target green circle, performance was impeded. This indicates that certain colors have an attention capturing effect even when they are irrelevant to the task at hand [19]. Further research has shown that features such as larger quantities, sharp contrast, curved lines, misaligned orientations, and non-prototypical colors or shapes are readily and often automatically detected. These features can be classified as exogenous attentional cues, as they have properties that automatically capture attention [20].

3 METHODOLOGY

3.1 Participants

Forty-six college students (17 males and 29 females, $M(\text{age}) = 20.44$, $SD(\text{age}) = 0.64$ years old) responded to our online post and registered for this experiment. All participants were right-handed, naïve to the specific purpose of this study, and had normal or corrected-to-normal vision and color vision. Seven individuals were excluded from the data analysis due to difficulties in adapting to motor control in the virtual environment, resulting in accuracy levels falling below chance. This led to a final sample size of 39 participants, including 15 males and 24 females. This study adhered to the ethical principles outlined in the Belmont Report [21] and received approval from the Institutional Human Research Ethics Committee for Non-Clinical Faculties. Prior to participation in the experiment, all participants provided written informed consent.

3.2 Experimental Design

Drawing inspiration from cognitive research on the sense of agency, we seek to accomplish two main objectives in our experiment. Firstly, we aim to explore the potential impact of manipulating prior control over a target on the speed of target detection during visual search in a virtual environment. Secondly, we intend to assess the degree to which the sense of agency serves as a protective factor against exogenous interference from salient attentional distractions, particularly within the framework of visual search tasks in a virtual environment.

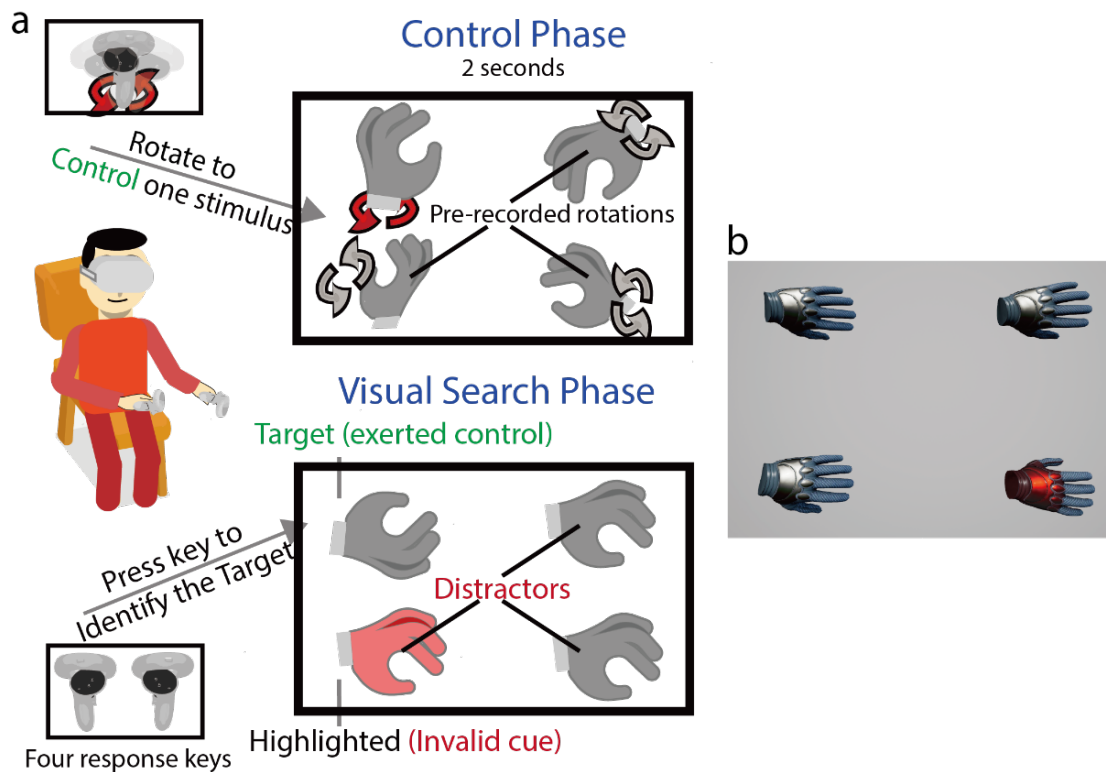


Figure 1: Illustration of (a) the experimental set up and procedures and (b) sample stimuli in the Visual Search Phase. During the control phase, participants were required to voluntarily control one hand-shaped stimulus while prerecorded rotations were shown for the other three hand-shaped stimulus. In the visual search phase, participants were tasked with identifying the target hand with thumb pointing upward. In this illustration, one of the distractor hands was highlighted in red to serve as a salient distraction in the visual search phase.

In pursuit of the aforementioned goals, we have elected to depart from the conventional method of collecting experimental responses via key press, and instead, have opted for the more ecologically valid approach of using a gyroscope within a gamepad to execute motor control to complete visual search within an immersive virtual reality setting. The strength of our design lies in its ability to specifically isolate the impact of agency. This was achieved by ensuring that only the controlled stimulus shared the movement onset with the participants' actions, allowing for a more accurate simulation of the sense of agency. Moreover, we have innovatively combined the realms of sense of agency and visual search to assess the robustness of the agency effect when confronted with compelling, attention-grabbing visual stimuli that vie for the cognitive resources of the individual in control. Here as below we present the details of the experimental settings.

The experiment comprised of two main phases.

Firstly, in the initial action **Control Phase** (see Figure 1a), a fixation cross was presented, splitting the screen into quarters. After one second, four virtual hands appeared. Participants were instructed to control one of the virtual hands by rotating their wrists, creating a sense of agency. The virtual hand they controlled moved in synchrony with their actions, while pre-recorded rotational movements were displayed for the other three virtual hands.

Participants were directed to identify the hand model they felt the most control over, with the understanding that this task was separate from the next phase of the experiment.

Two seconds following the initial action control phase, the **Visual Search Phase** started with the appearance of four static hand models, wherein only the designated target hand differed from the distractors (see Figure 1b). Participants were required to promptly and accurately differentiate the target hand from the others (thumb pointing upward rather than downward in Figure 1b) by pressing spatially congruent buttons on the controller. The reaction time was recorded from the initiation of the second phase to the button press. To introduce exogenous interference in the visual search task, we manipulated the color tint of either the target hand (valid attentional cue) or one of the distractor hands (invalid attentional cue) to red.

In sum, the experiment involved two factors, each with two levels: whether the virtual hand participant controlled in the initial action control phase would later become the search target in the visual search phase (Control target) or not (Control distractor), and whether the stimulus highlighted in the visual search phase was the search target (Valid attentional cue) or not (Invalid attentional cue). Throughout the whole experiment, the display remained fixed, ensuring that the rendered image did not track the movement of

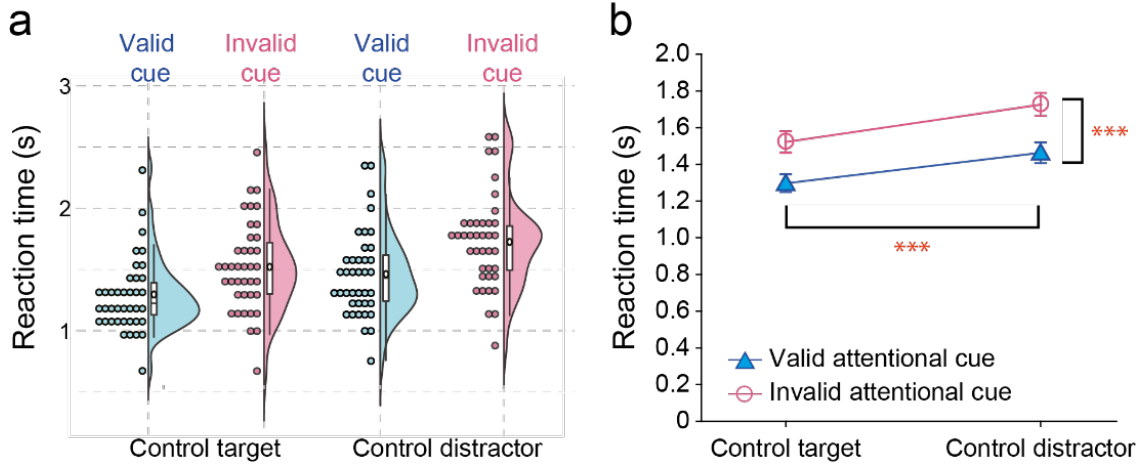


Figure 2: (a) Distribution of reaction time of all participants in four different conditions. Each dot indicates one participant and the distribution shape indicates the probability density of the data at different values. (b) Participants identified the search target faster when they had previously controlled the searched target and when the visually prominent cue was valid. *, $p < .001$**

the head, thus maintaining a consistent relative position with the eyes.

3.3 Procedure

Before commencing the experiment, all participants received guidance from an instructional video detailing the experimental procedure. The experiment consisted of a total of 64 trials, with each trial comprising of two phases. The order of these trials was completely randomized. The entire experiment took around 10 minutes to complete.

3.4 Apparatus

We conducted the experiment in a virtual reality setting using an Oculus Quest 2 headset (Meta Platform Inc, U. S., refresh rate: 72Hz, horizontal field of view: $89^\circ \pm 4^\circ$, vertical field of view: $93^\circ \pm 5.1^\circ$). We made use of Unreal engine 5.2.1 [22] to construct the virtual environment, with programs running on a computer equipped with an Intel Core i9-13900T processor and Nvidia RTX 4070 graphics card. The content was streamed to the headset via the Oculus Home app 56.0.0.76.155, along with a USB3.2 gen2 cable.

3.5 Data Processing

Performance in the visual search task was measured using two metrics: reaction time (the time from the onset of visual search to the key press) and accuracy (the number of correct trials divided by all trials). Typically, faster reaction times indicate better performance, assuming accuracy remains constant. In cases where there is no significant difference in accuracy among conditions, the mean reaction time values can be used to assess performance.

We used a within-subject design in this study and thus utilized repeated measures Analysis of Variance (ANOVA) to investigate the effect of sense of agency (Control Target; Control Distractor) and exogenous salient cue (Valid Attentional Cue; Invalid Attentional Cue) on visual search performance.

4 RESULTS

In the initial **Control Phase**, all participants reported they could effortlessly find the controlled hand. In the **Visual Search Phase**, the accuracy was consistently high across all conditions, with results as follows: Control target $\times$ Valid attentional Cue (Mean $\pm$ SD: 0.96 ± 0.06); Control target $\times$ Invalid attentional Cue (0.94 ± 0.12); Control distractor $\times$ Valid attentional Cue (0.95 ± 0.09); Control distractor $\times$ Invalid attentional Cue (0.94 ± 0.10). We excluded outlier trials, defined as those falling more than 2.5 standard deviations outside the mean reaction time for each participant, which represented 4.13% of the total trials. Below we report the results of the reaction time in the **Visual Search Phase**.

Figure 2a displays the reaction time distributions across four conditions. A 2 (Cue validity: Valid attentional Cue vs. Invalid attentional Cue) $\times$ 2 (Sense of agency: Control target vs. Control distractor) ANOVA revealed significant main effects for both cue validity, $F(1, 38) = 46.05$, $p < .001$, $\eta^2 = .54$, indicating that the visually salient cues effectively directed participants' attention to the highlighted stimulus, and for the sense of agency, $F(1, 38) = 26.72$, $p < .001$, $\eta^2 = .41$, showing a consistent preference for the stimulus that participants controlled (Figure 2b). Specifically, regardless of the validity of the cue, the time it took to identify the search target and press the button was 11.32% faster if the search target was the controlled virtual hand during the **initial Control phase** (Mean $\pm$ SD: $1.41 \pm .35$ s) compared to when it was not ($1.59 \pm .39$ s). This result supports the hypothesis proposed in **RQ1**, affirming that modifying previous control over a target accelerates the target identification during subsequent visual search in a virtual environment.

However, we did not observe a significant interaction effect between cue validity and the sense of agency ($F(1, 38) = 1.31$, $p = .26$, $\eta^2 = .03$). When considering the effect sizes, the influence of attentional cue ($\eta^2 = .54$) was more substantial than that of the sense of agency ($\eta^2 = .41$). This suggests that manipulating attention

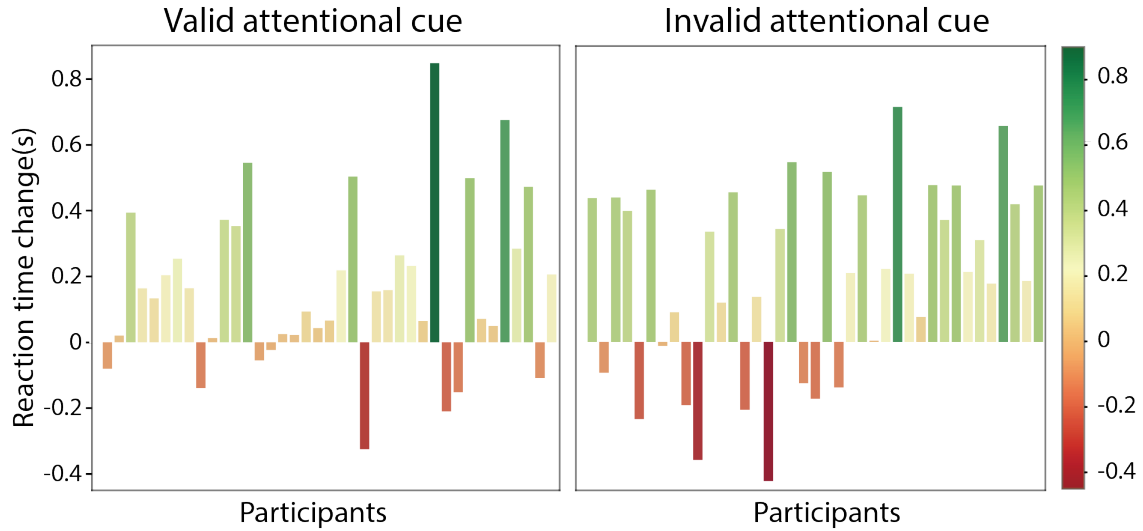


Figure 3: Bar chart illustrating the individual differences in reaction time changes, calculated as reaction time difference between the Control target and Control distractor conditions. Each bar represents an individual participant and the color bar on the right offers a visual aid for interpreting the reaction time changes.

through salient cues may have a stronger impact on participants' visual search performance than the sense of agency, which provides insights into the interplay between sense of agency and exogenous information addressed in **RQ2**.

Last, our data revealed notable variations among individuals in their responsiveness to the sense of agency, as depicted in Figure 3. Specifically, we found that around eight and ten out of thirty-nine participants exhibited a delay in identifying the search target in the two attentional cue conditions respectively, even though they had previously demonstrated control over the same target during the Control phase of the experiments. This highlights the complexity of human agency and the variability in how people perceive and respond to their sense of agency.

5 DISCUSSION

5.1 The Role of Sense of Agency in Visual Search

Regarding **RQ1**, our research revealed that participants were more efficient in identifying and responding to a target when they had a prior control of a virtual hand located at the same spatial position of the target. The promotional effect of sense of agency could be explained by human instinct of highlighting the process of stimuli within our control, to learn the action-effect associations [1]. Even though rationally, attending to a possibly misleading cue could accompany with a larger cost of time and accuracy, participants in this study illustrated an overall inclination toward stimulus with more sense of agency [8]. Additionally, we observed that even when presented with distracting cues, the sense of agency remained influential in visual search. This answers **RQ2** and suggests that the sense of agency plays a significant role in how individuals prioritize and interact with stimuli, even in the presence of distractions.

5.2 Individual Variations in the Sense of Agency

By closely analyzing the reaction time disparity between the trials where participants exerted prior control over the virtual hand in the same spatial location of the target in the Visual Search phase and those where they did not, we noted that approximately 21%-26% of the participants exhibited a contrasting pattern to the anticipated facilitation effect of the sense of agency. This suggests that individuals vary in their receptiveness to the sense of agency, and a sizable portion of the population does not experience the advantage from the sense of agency for a subsequent task that occurs after exerting control. Here as below we listed several potential explanations for this observation.

Firstly, we conducted 16 trials for each condition, with completely randomized presentation. Due to the limited number of trials, some participants may have experienced consecutive occurrence of same type of trial, leading to an over estimation of the connection between the self-control of the virtual hand in the initial Control phase and the target in the second Visual Search phase. As a result, they may have expected search target to appear in certain locations with higher frequency than others [17], leading to a decrease in the proportion of allocated attention to the exogenous attentional cue. To address this issue, we can use a larger number of trials and vary the contingency gap between the first and second experimental phases in our next investigation.

Secondly, individuals may interpret experimental instructions differently, and these subjective interpretations can influence their behavior during the experiment. For example, coercive experimental instructions have been shown to discourage the sense of agency and the intention to actively engage in the task [23]. The fluency of control is not the sole determinant of the sense of agency; the intention to control also plays a significant role [24]. If the intention to concentrate varies, the sense of agency and resulting performance are likely to change. In our experiment, we mitigated

these instructional effects by asking participants to perform wrist rotational movements at their own pace. Nevertheless, we still observed diverse individual approaches to utilizing the sense of agency [25], with explorative participants scattering their focus to observe how uncontrolled stimuli change, while exploitative participants tended to remain focused on the controlled stimulus. A subsequent user study with surveys could provide further insight into this phenomenon.

6 FUTURE WORK AND IMPLICATIONS

Sense of agency is an indispensable part of daily interaction with the external world [3]. It helps people to filter out self-unrelated information in an information abundant environment. It is also a crucial ability of leading people to learn a new skill by reducing prediction error through practice. By creatively collaborating with technologies, our findings should bring insights into design principles of human-computer interaction.

6.1 Research Gaps and Future Investigations

We investigated how the sense of agency can maintain attention amid visual distractions and suggested that future research could expand on our findings in diverse ways. For instance, future studies could use augmented reality and eye-tracking to study initial eye movements and attention capture in more depth. Research could also explore how individual differences in the sense of agency within virtual environments affect visual search tasks and the relationship between perceived control and attentional efficiency. Finally, the impact of endogenous cues, like directional arrows, on attentional control and the sense of agency in visual search tasks warrants further investigation.

6.2 Implications in Human-computer Interaction Design Principles

We believe that our findings are prospective for design principles in user guidance, especially for users lack first-hand experience with a technology. For instance, active interaction for tutorial design in virtual reality is as important as verbal and visual guidance. Importantly, as observed in our results, individual variations exist in sense of agency. Thus, satisfying curiosity of explorative learners and offer accurate feedback for exploitative learners can be two facets of a useful design.

Ensuring focused attention is crucial for a positive user experience, which aligns closely with our research findings. In practical terms, consider the case of notifications, which are a common physical interruption. Current designs for apps and operating systems typically offer a binary choice with "do not disturb" settings, leading users to either endure unnecessary interruptions or overlook important, task-relevant messages. Future devices could improve this by selectively filtering notifications, taking the user's responsiveness into account to the sense of agency. For instance, understanding how the sense of agency differs between novice and experienced users can lead to interface designs that better accommodate the needs of both groups.

7 CONCLUSION

In conclusion, our findings suggest that prior control over a target would accelerate the detection of that target during later visual search in a virtual environment, and the improvement survived external distraction from physically salient information, highlighting the importance of a sense of agency in information processing. These findings serve as a solid base for future research aimed at delving deeper into the complex interactions between sense of agency and visual attention.

The implications of this research are broad, offering valuable insights for the design of user interfaces in virtual reality, particularly in educational and training settings. Designers can leverage the sense of agency to facilitate quicker and more efficient learning experiences, tailoring interactions to cater to both explorative and exploitative learning styles. Furthermore, interface designers should take users' responsiveness to the sense of agency into consideration to optimize digital experiences.

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